

Generalization of Marion's Theorem: Volumes of Central Polytopes Obtained by Trisecting the Edges of Simplices

Yu. V. Kazakov, Cand. Sci. (Phys.-Math.)*

Reshetnev Siberian State University of Science and Technology(retired)
Address: 31, Krasnoyarskii Rabochii Ave., 82, Mira Ave., 660037, Krasnoyarsk, Russia

Abstract

The classical Marion's theorem states that the area of the central hexagon obtained by dividing each side of a triangle into three equal parts and connecting the interior division points is exactly $1/10$ of the area of the original triangle. In this paper, the construction is extended to an arbitrary n -dimensional simplex: each edge is divided into three equal parts, and through each $n-1$ vertices and the two interior division points of the opposite edge, two hyperplanes are drawn. It is shown that the resulting inner polytope has a surprisingly simple volume formula: the volume is $1/\binom{2n+1}{n}$ times the volume of the original simplex. The proof uses a probabilistic interpretation with exponential random variables and reduces the geometric problem to a combinatorial binomial sum. Additionally, the combinatorics of the polytope are discussed: the numbers of vertices, edges, and faces are expressed in closed form, and the complete f -vector is described. A link to an interactive three-dimensional illustration for a regular tetrahedron in GeoGebra [5] is provided. For a clearer understanding of the solution approach, Appendix A presents a detailed analysis of the probabilistic method using the triangle as an example, Appendix B presents the technique of applying barycentric coordinates to solve the problem in the three-dimensional case, and Appendix C contains Asymptote [10] code for visualizing the central deltoidal dodecahedron inside a regular tetrahedron.

1 Introduction

Marion's theorem [1, 2] is an elegant elementary result in plane geometry: in any triangle, the hexagon formed by the interior points dividing each side in the ratio $1 : 2$ has an area exactly one tenth of the area of the whole triangle (Fig. 1). Through each vertex of the triangle, two lines are drawn to the two trisection points of the opposite side; the region between these lines in the center of the triangle forms a hexagon. The area fraction $1/10$ of this hexagon is independent of the shape of the triangle.

Generalizations of this construction to higher dimensions are absent in the literature — the author first addressed this problem in 2009 and obtained a result for the case of

*E-mail: yu.v.kazakov@gmail.com

trisecting the edges of a tetrahedron in 3D using Mathcad. In three-dimensional space, an analogous procedure applied to a tetrahedron (each edge is divided into three equal parts, planes are drawn through two vertices and an interior point on the opposite edge — 12 planes in total) yields a central polytope whose volume is $1/35$ of the volume of the tetrahedron [4]. A live model demonstrating this construction for a regular tetrahedron is available on the GeoGebra website [5]; the central polytope in this case is a deltoidal dodecahedron (a name proposed by the author of this note, since the faces of the polytope are identical deltoids, and the polytope is analogous to the rhombic dodecahedron). In four-dimensional space, the volume fraction turns out to be $1/126$, where for each set of 3 vertices, two hyperplanes are constructed, each containing one of the two points on the opposite edge (the opposite edge is also divided into three equal parts).

The obtained volume fractions of the inner polytopes —

$$\frac{1}{10} = \frac{1}{\binom{5}{2}}, \quad \frac{1}{35} = \frac{1}{\binom{7}{3}}, \quad \frac{1}{126} = \frac{1}{\binom{9}{4}}$$

— unambiguously point to a general formula:

$$\frac{\text{Vol}(\text{inner polytope})}{\text{Vol}(\text{simplex})} = \frac{1}{\binom{2n+1}{n}}.$$

The purpose of this paper is to provide a rigorous proof of this formula for arbitrary dimension $n \geq 1$ and to describe the combinatorial structure of the resulting central polytope.

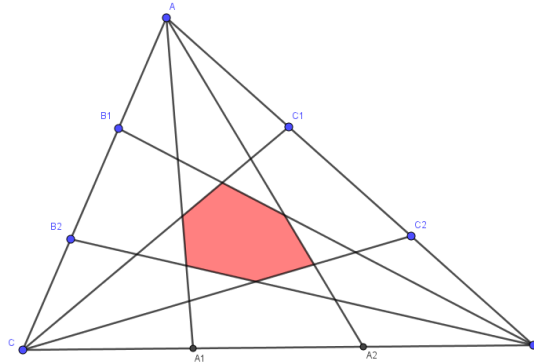


Figure 1: Marion's central hexagon in a triangle. Image taken from [6].

2 Construction and Inequalities

Consider an n -dimensional simplex Δ^n . In barycentric coordinates $(x_0, x_1, \dots, x_n)$, it is defined by the conditions

$$x_i \geq 0, \quad \sum_{i=0}^n x_i = 1.$$

The ratio of the volume of the inner polytope to the volume of the original simplex is an affine invariant; therefore, without loss of generality, we can use this barycentric representation. The volume of the standard simplex in this normalization is $V_0 = 1/n!$,

but the specific volume value is irrelevant, since only the volume ratio enters the final result.

Let $\mathbf{e}_0, \mathbf{e}_1, \dots, \mathbf{e}_n$ be the vertices of the simplex. In barycentric coordinates, the vertex $\mathbf{e}_i$ has coordinates $(x_0, \dots, x_n)$ with $x_i = 1$ and $x_j = 0$ for $j \neq i$.

Choose an arbitrary edge of the simplex corresponding to vertices $\mathbf{e}_i$ and $\mathbf{e}_j$ ($i \neq j$). The two points dividing this edge internally in the ratios $1 : 2$ and $2 : 1$ have barycentric coordinates

$$P_{ij} = \frac{2}{3}\mathbf{e}_i + \frac{1}{3}\mathbf{e}_j, \quad Q_{ij} = \frac{1}{3}\mathbf{e}_i + \frac{2}{3}\mathbf{e}_j.$$

Consider all $n - 1$ vertices other than $\mathbf{e}_i$ and $\mathbf{e}_j$. A hyperplane passes through these $n - 1$ vertices and the point P_{ij} ; a similar hyperplane passes through the same $n - 1$ vertices and Q_{ij} . In barycentric coordinates, these hyperplanes correspond to the equations

$$x_j = 2x_i \quad \text{and} \quad x_i = 2x_j.$$

The half-spaces containing the central region are described by the inequalities

$$\frac{1}{2}x_j \leq x_i \leq 2x_j.$$

Repeating this construction for each edge of the simplex, we obtain for any ordered pair $0 \leq i, j \leq n$, $i \neq j$, the constraint

$$x_i \leq 2x_j. \tag{1}$$

The inner polytope $\mathcal{P}_n$ is defined as the set of points satisfying

$$x_i \geq 0, \quad \sum_{i=0}^n x_i = 1, \quad x_i \leq 2x_j \quad \text{for all } i \neq j.$$

By symmetry, it suffices to require the inequalities $x_i \leq 2x_j$ for all $i \neq j$; the reverse inequalities $x_j \leq 2x_i$ then hold automatically, as they are obtained by permuting indices.

The polytope $\mathcal{P}_n$ is the intersection of the original simplex with a system of half-spaces bounded by the hyperplanes

$$x_i = 2x_j \quad (i \neq j).$$

As will be shown below, it is the convex hull of $2^{n+1} - 2$ points whose barycentric coordinates take only two distinct values differing by a factor of two.

3 Volume Calculation

To find the volume of $\mathcal{P}_n$, we employ a probabilistic method.

Let $Y_0, Y_1, \dots, Y_n$ be independent exponentially distributed random variables with mean 1. Then the random vector

$$\left(\frac{Y_0}{S}, \frac{Y_1}{S}, \dots, \frac{Y_n}{S} \right), \quad S = \sum_{k=0}^n Y_k,$$

is uniformly distributed on the simplex Δ^n [7]. Consequently,

$$\frac{\text{Vol}(\mathcal{P}_n)}{\text{Vol}(\Delta^n)} = \mathbb{P}(\text{the conditions } x_i \leq 2x_j \text{ hold}) = \mathbb{P}(Y_i \leq 2Y_j \text{ for all } i, j).$$

The event $Y_i \leq 2Y_j$ for all i, j means that the ratio of the minimum to the maximum among Y_k is at least $1/2$. By symmetry, we fix which variable is the smallest. Let Y_0 be the minimum; then the condition becomes $Y_k \leq 2Y_0$ for all $k = 1, \dots, n$ (since $Y_0 \leq Y_k$ automatically implies $Y_0 \leq 2Y_k$). The probability that Y_0 is the minimum and that $Y_k \leq 2Y_0$ for $k \geq 1$ is

$$\int_0^\infty e^{-y_0} \left(\int_{y_0}^{2y_0} e^{-y} dy \right)^n dy_0 = \int_0^\infty e^{-y_0} (e^{-y_0} - e^{-2y_0})^n dy_0.$$

Multiplying by $n+1$ (the number of choices for which variable is the minimum), we obtain

$$\frac{\text{Vol}(\mathcal{P}_n)}{V_0} = (n+1) \int_0^\infty e^{-y_0} (e^{-y_0} - e^{-2y_0})^n dy_0 = (n+1) \int_0^\infty e^{-(n+1)y_0} (1 - e^{-y_0})^n dy_0.$$

Expanding $(1 - e^{-y_0})^n$ using the binomial theorem:

$$(1 - e^{-y_0})^n = \sum_{k=0}^n \binom{n}{k} (-1)^k e^{-ky_0}.$$

The integral becomes a sum of elementary exponential integrals:

$$\begin{aligned} \frac{\text{Vol}(\mathcal{P}_n)}{V_0} &= (n+1) \sum_{k=0}^n \binom{n}{k} (-1)^k \int_0^\infty e^{-(n+1+k)y_0} dy_0 \\ &= (n+1) \sum_{k=0}^n \binom{n}{k} \frac{(-1)^k}{n+1+k}. \end{aligned}$$

A known combinatorial identity (see, e.g., [8]) states that

$$\sum_{k=0}^n \binom{n}{k} \frac{(-1)^k}{m+k} = \frac{1}{m \binom{m+n}{n}} \quad (m > 0).$$

Setting $m = n+1$, we find

$$\sum_{k=0}^n \binom{n}{k} \frac{(-1)^k}{n+1+k} = \frac{1}{(n+1) \binom{2n+1}{n}}.$$

Multiplying by $n+1$, we arrive at the main result

$$\boxed{\frac{\text{Vol}(\mathcal{P}_n)}{\text{Vol}(\Delta^n)} = \frac{1}{\binom{2n+1}{n}}}. \quad (2)$$

Thus, the volume of the inner polytope is exactly $1/\binom{2n+1}{n}$ times the volume of the original simplex. For $n = 1$ (a segment) we get $1/3$, for $n = 2$ — $1/10$, for $n = 3$ — $1/35$ — these are simple cases. For $n = 4$, the formula yields $1/126$, which agrees with Python calculations for the volume of the inner convex hull (volume calculations were performed using both rigorous geometric approaches and the Monte Carlo probabilistic method).

4 Combinatorial Structure

In addition to the volume, the inner polytope $\mathcal{P}_n$ has a rich combinatorial structure that can be described through the system of hyperplanes $x_i = 2x_j$.

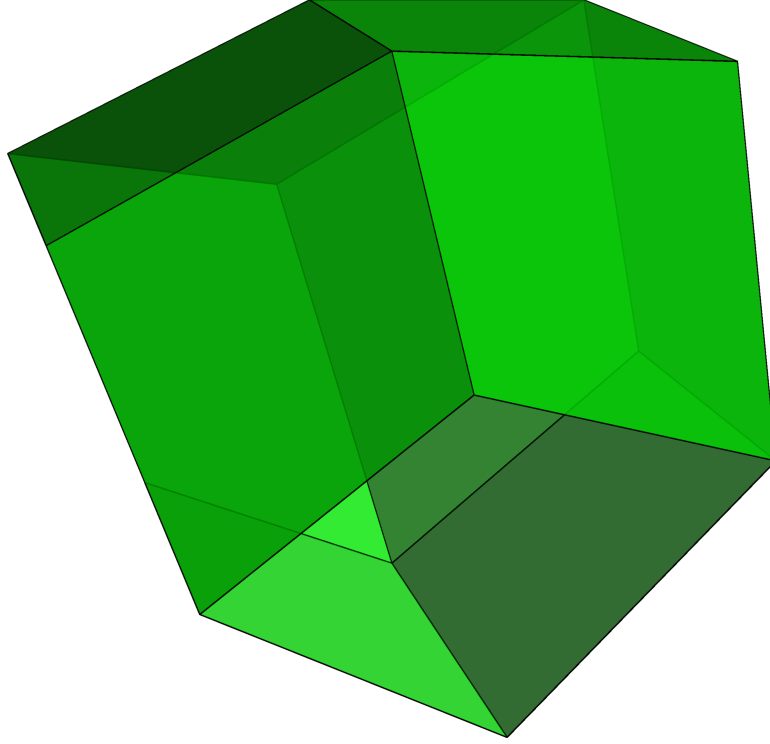


Figure 2: Central polytope obtained in a three-dimensional tetrahedron (deltoidal dodecahedron).

4.1 Vertices

A vertex of $\mathcal{P}_n$ occurs when the maximum number of constraints $x_i \leq 2x_j$ is active. It is easy to verify that the vertices are precisely those points whose barycentric coordinates take only two distinct values, say α and 2α , with the number of coordinates equal to α being k , and the number of coordinates equal to 2α being $n+1-k$ ($1 \leq k \leq n$). From the condition $\sum x_i = 1$, we find $\alpha = 1/(2n+2-k)$ or $\alpha = 1/(n+1+k)$, depending on which value appears k times. Since the roles of α and 2α can be interchanged, each nontrivial partition of the $n+1$ coordinates into two nonempty sets yields two vertices. Thus, the total number of vertices is

$$f_0 = 2(2^n - 1) = 2^{n+1} - 2.$$

4.2 Edges

Two vertices are connected by an edge if their partitions differ by moving exactly one coordinate from the set with value α to the set with value 2α (or vice versa) while preserving the overall ratio of values $1/2$ or 2 . A vertex corresponding to a partition of sizes $(k, n+1-k)$ has degree $k + (n+1-k) = n+1$ for $2 \leq k \leq n-1$; for $k=1$ or $k=n$, the degree is n , since moving a coordinate that would make one of the sets empty is forbidden. Summing the degrees over all vertices and dividing by 2, we find the number of edges:

$$\begin{aligned} f_1 &= \frac{1}{2} \left(2 \cdot n \cdot \binom{n+1}{1} \right) \quad (\text{for } k=1 \text{ and } k=n) + \sum_{k=2}^{n-1} 2 \binom{n+1}{k} (n+1) \\ &= (n+1)(2^n - 2). \end{aligned}$$

The polytope $\mathcal{P}_n$ is a hypersimplex truncated by additional half-spaces $x_i \leq 2x_j$, and its graph is a subgraph of the Johnson graph.

4.3 Faces

Each hyperface (face of codimension 1) corresponds to an equality $x_i = 2x_j$ for some ordered pair (i, j) , $i \neq j$. There are $n(n+1)$ such ordered pairs, and each inequality $x_i \leq 2x_j$ is essential (removing it changes the polytope). Consequently, the number of hyperfaces is

$$f_{n-1} = n(n+1).$$

Each hyperface is combinatorially equivalent to an $(n-1)$ -dimensional cube.

The complete f -vector can be obtained from the theory of hyperplane arrangements (see [9]) or via generating functions for face numbers; the polytope $\mathcal{P}_n$ is combinatorially isomorphic to the so-called *shard polytope* associated with $\mathbf{c}$ -vectors and dual to the associahedron.

5 Conclusion and Perspectives

We have shown that the volume fraction of the central polytope obtained by trisecting all edges of an n -dimensional simplex and cutting off the corresponding hyperplanes is exactly

$$\frac{\text{Vol}(\text{inner polytope})}{\text{Vol}(\text{simplex})} = \frac{1}{\binom{2n+1}{n}}.$$

This generalizes Marion's theorem for triangles ($n = 2$) to tetrahedra ($n = 3$) and simplices of higher dimensions. The proof, based on a probabilistic representation of the uniform distribution on the simplex, transforms the geometric problem into a simple integral and binomial sum. The same method immediately yields combinatorial data: the number of vertices is $2^{n+1} - 2$, edges are $(n+1)(2^n - 2)$, and hyperfaces are $n(n+1)$.

Several open questions remain. One could investigate the volume for an arbitrary integer division ratio r instead of 3; the probability then leads to a sum of the form $(n+1) \sum_{k=0}^n \binom{n}{k} (-1)^k / (n+1+k(r-1))$, which simplifies to a rational number but, apparently, does not reduce to a single binomial coefficient. It would also be interesting to study the dual polytope and its volume, or to consider the limiting shape as $n \rightarrow \infty$ after appropriate scaling.

A Probabilistic Method Using the Triangle as an Example

In this appendix, we reproduce step by step the probabilistic calculation of the area of Marion's central hexagon (the case $n = 2$) to illustrate the general method presented in the paper.

Triangle in Barycentric Coordinates

Consider the standard 2-simplex (triangle) with vertices $\mathbf{e}_0 = (1, 0, 0)$, $\mathbf{e}_1 = (0, 1, 0)$, $\mathbf{e}_2 = (0, 0, 1)$. Any point of the triangle is given by coordinates (x_0, x_1, x_2) with $x_i \geq 0$, $x_0 + x_1 + x_2 = 1$.

Marion's construction: on each side we choose two points dividing it into three equal parts. Through each vertex and the two division points on the opposite side, we draw two lines. All six lines bound the central hexagon $\mathcal{P}_2$. In barycentric coordinates, these lines correspond to the equations

$$x_1 = 2x_2, x_2 = 2x_1 \quad (\text{for edge } \mathbf{e}_0\mathbf{e}_1 \text{ and vertex } \mathbf{e}_2),$$

and similarly for the other edges. The interior region is defined by the inequalities $x_i \leq 2x_j$ for all $i \neq j$.

Uniform Distribution on the Triangle

The key idea: if Y_0, Y_1, Y_2 are independent random variables with the standard exponential distribution (density e^{-y} , $y \geq 0$), then the vector

$$(X_0, X_1, X_2) = \left(\frac{Y_0}{S}, \frac{Y_1}{S}, \frac{Y_2}{S} \right), \quad S = Y_0 + Y_1 + Y_2,$$

is uniformly distributed in the standard triangle. This is a well-known fact from probability theory (the inverse transform method for the Dirichlet distribution). Therefore, the probability that a random point falls in the hexagon $\mathcal{P}_2$ equals the ratio of areas:

$$\frac{\text{Area}(\mathcal{P}_2)}{\text{Area}(\Delta^2)} = \mathbb{P}(X_i \leq 2X_j \ \forall i \neq j) = \mathbb{P}(Y_i \leq 2Y_j \ \forall i, j).$$

Probability Calculation

The event $Y_i \leq 2Y_j$ for all i, j means that the ratio of the maximum to the minimum of the variables does not exceed 2. We use symmetry: the probability equals three times the probability that Y_0 is the smallest of the three and that $Y_1 \leq 2Y_0, Y_2 \leq 2Y_0$.

Thus, we compute

$$P = 3 \int_0^\infty f_{Y_0}(y_0) \mathbb{P}(Y_1 \leq 2y_0 \mid Y_1 \geq y_0) \mathbb{P}(Y_2 \leq 2y_0 \mid Y_2 \geq y_0) dy_0,$$

where the condition $Y_i \geq y_0$ follows from y_0 being the minimum. The density of Y_0 is e^{-y_0} , and

$$\mathbb{P}(Y_1 \in [y_0, 2y_0]) = \int_{y_0}^{2y_0} e^{-y} dy = e^{-y_0} - e^{-2y_0}.$$

Hence,

$$P = 3 \int_0^\infty e^{-y_0} (e^{-y_0} - e^{-2y_0})^2 dy_0.$$

Transform the integrand:

$$e^{-y_0} (e^{-y_0} - e^{-2y_0})^2 = e^{-y_0} \cdot e^{-2y_0} (1 - e^{-y_0})^2 = e^{-3y_0} (1 - e^{-y_0})^2.$$

Thus,

$$P = 3 \int_0^\infty e^{-3y_0} (1 - e^{-y_0})^2 dy_0.$$

Expand the square: $(1 - e^{-y_0})^2 = 1 - 2e^{-y_0} + e^{-2y_0}$. The integral becomes a sum:

$$\begin{aligned} P &= 3 \int_0^\infty e^{-3y_0} (1 - 2e^{-y_0} + e^{-2y_0}) dy_0 \\ &= 3 \left[\int_0^\infty e^{-3y_0} dy_0 - 2 \int_0^\infty e^{-4y_0} dy_0 + \int_0^\infty e^{-5y_0} dy_0 \right] \\ &= 3 \left[\frac{1}{3} - 2 \cdot \frac{1}{4} + \frac{1}{5} \right] \\ &= 3 \left(\frac{1}{3} - \frac{1}{2} + \frac{1}{5} \right) \\ &= 3 \cdot \frac{10 - 15 + 6}{30} = 3 \cdot \frac{1}{30} = \frac{1}{10}. \end{aligned}$$

Thus, the area of the hexagon is $1/10$ of the area of the original triangle.

Connection with the General Formula

For $n = 2$, the general formula gives $1/\binom{5}{2} = 1/10$, which coincides with the obtained result. Note that the sum $1/3 - 2/4 + 1/5$ is equivalent to the expression $(n+1) \sum_{k=0}^n \frac{\binom{n}{k} (-1)^k}{n+1+k}$ for $n = 2$, and after simplification yields the same value.

This example clearly demonstrates how the probabilistic representation reduces a geometric problem to simple integration and algebraic combinatorics. Calculations for higher dimensions proceed analogously, where the power $(1 - e^{-y_0})^n$ generates the binomial coefficient $\binom{2n+1}{n}$ in the denominator.

B Volume Calculation of the Inner Polytope for a Tetrahedron. Parameters of the Deltoid

In this appendix, we obtain the central deltoidal dodecahedron for the tetrahedron with vertices $(0, 0, 0)$, $(1, 1, 0)$, $(1, 0, 1)$, $(0, 1, 1)$ by direct geometric computation, and find its volume and face parameters — congruent deltoids (kites).

Coordinates and Cutting Planes

Consider the tetrahedron with vertices (in three-dimensional space):

$$\begin{aligned} A &= (0, 0, 0), \\ B &= (1, 1, 0), \\ C &= (1, 0, 1), \\ D &= (0, 1, 1). \end{aligned}$$

Conversion Formula from Barycentric to Cartesian Coordinates

For this tetrahedron, the transformation matrix from barycentric coordinates (x_A, x_B, x_C, x_D) to Cartesian coordinates (x, y, z) is:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} x_A \\ x_B \\ x_C \\ x_D \end{pmatrix}.$$

In expanded form:

$$\begin{cases} x = x_B + x_C, \\ y = x_B + x_D, \\ z = x_C + x_D. \end{cases}$$

Vertices of the Deltoidal Dodecahedron

According to the general theory, the vertices of the polytope are points where the coordinates take only two values: α and 2α , with $k\alpha + (4 - k)2\alpha = 1$, where k is the number of coordinates equal to α . We obtain:

- $k = 1$: $\alpha = 1/7$, $2\alpha = 2/7$ (4 vertices);
- $k = 2$: $\alpha = 1/6$, $2\alpha = 1/3$ (6 vertices);
- $k = 3$: $\alpha = 1/5$, $2\alpha = 2/5$ (4 vertices).

Total: $2^4 - 2 = 14$ vertices.

No.	Letter	Barycentric coordinates	Cartesian coordinates
1	A	$(2/7, 1/7, 2/7, 2/7)$	$(3/7, 3/7, 4/7)$
2	B	$(2/7, 2/7, 1/7, 2/7)$	$(3/7, 4/7, 3/7)$
3	C	$(2/7, 2/7, 2/7, 1/7)$	$(4/7, 3/7, 3/7)$
4	D	$(1/7, 2/7, 2/7, 2/7)$	$(4/7, 4/7, 4/7)$
5	E	$(1/3, 1/6, 1/6, 1/3)$	$(1/3, 1/2, 1/2)$
6	F	$(1/3, 1/6, 1/3, 1/6)$	$(1/2, 1/3, 1/2)$
7	G	$(1/3, 1/3, 1/6, 1/6)$	$(1/2, 1/2, 1/3)$
8	H	$(1/6, 1/3, 1/6, 1/3)$	$(1/2, 2/3, 1/2)$
9	I	$(1/6, 1/3, 1/3, 1/6)$	$(2/3, 1/2, 1/2)$
10	J	$(1/6, 1/6, 1/3, 1/3)$	$(1/2, 1/2, 2/3)$
11	K	$(2/5, 1/5, 1/5, 1/5)$	$(2/5, 2/5, 2/5)$
12	L	$(1/5, 2/5, 1/5, 1/5)$	$(3/5, 3/5, 2/5)$
13	M	$(1/5, 1/5, 2/5, 1/5)$	$(3/5, 2/5, 3/5)$
14	N	$(1/5, 1/5, 1/5, 2/5)$	$(2/5, 3/5, 3/5)$

Table 1: Vertices of the deltoidal dodecahedron in barycentric and Cartesian coordinates.
Vertices 1–4: $k = 1$ (4 vertices), 5–10: $k = 2$ (6 vertices), 11–14: $k = 3$ (4 vertices).

Table of Vertices of the Deltoidal Dodecahedron

Explanation of the table:

- Vertices with $k = 1$ (No. 1–4): one coordinate equals $1/7$, three coordinates equal $2/7$.
- Vertices with $k = 2$ (No. 5–10): two coordinates equal $1/6$, two coordinates equal $1/3$.
- Vertices with $k = 3$ (No. 11–14): three coordinates equal $1/5$, one coordinate equals $2/5$.
- Cartesian coordinates are obtained by $x = x_B + x_C$, $y = x_B + x_D$, $z = x_C + x_D$.

Volume of the Deltoidal Dodecahedron

The volume of the original tetrahedron with vertices $(0, 0, 0)$, $(1, 1, 0)$, $(1, 0, 1)$, $(0, 1, 1)$ is:

$$V_{\text{tetra}} = \frac{1}{3}.$$

The volume of the deltoidal dodecahedron can be found by the general formula:

$$V_{\text{del}} = \frac{V_{\text{tetra}}}{\binom{7}{3}} = \frac{1}{35} \cdot \frac{1}{3} = \frac{1}{105}.$$

Alternative Calculation of the Volume Fraction via Determinant

The volume fraction of the inner polytope can also be calculated using the fact that the vertices of the polytope are convex combinations of the simplex vertices. For more details on barycentric coordinates, see the article on MathWorld [3].

For the central polytope in a tetrahedron, the volume can be computed via a determinant composed of the barycentric coordinates of four points: three vertices of one face and the center. The determinant

$$24 \cdot \det \begin{pmatrix} 2/7 & 1/7 & 2/7 & 2/7 \\ 1/3 & 1/6 & 1/6 & 1/3 \\ 2/5 & 1/5 & 1/5 & 1/5 \\ 1/4 & 1/4 & 1/4 & 1/4 \end{pmatrix} = \frac{1}{35}$$

gives the volume fraction of the inner polytope. The factor 24 arises because each of the 12 deltoids is divided into two identical triangles, and thus the inner polytope is composed of 24 identical pyramids.

Geometry of the Deltoid

The faces of the deltoidal dodecahedron are 12 congruent deltoids. Consider one face corresponding to the equality $x_A = 2x_B$. Its vertices (in barycentric coordinates) are:

$$\begin{aligned} V_1 &= (2/7, 1/7, 2/7, 2/7), \\ V_2 &= (1/3, 1/6, 1/6, 1/3), \\ V_3 &= (1/3, 1/6, 1/3, 1/6), \\ V_4 &= (2/5, 1/5, 1/5, 1/5). \end{aligned}$$

After converting to Cartesian coordinates using the formulas $x = x_B + x_C$, $y = x_B + x_D$, $z = x_C + x_D$, we obtain:

$$\begin{aligned} V_1 &\rightarrow \left(\frac{3}{7}, \frac{3}{7}, \frac{4}{7}\right), \\ V_2 &\rightarrow \left(\frac{1}{3}, \frac{1}{2}, \frac{1}{2}\right), \\ V_3 &\rightarrow \left(\frac{1}{2}, \frac{1}{3}, \frac{1}{2}\right), \\ V_4 &\rightarrow \left(\frac{2}{5}, \frac{2}{5}, \frac{2}{5}\right). \end{aligned}$$

Computing the side lengths and diagonals:

- Short sides:

$$|V_1V_2| = |V_1V_3| = \frac{\sqrt{34}}{42}$$

- Long sides:

$$|V_2V_4| = |V_3V_4| = \frac{\sqrt{22}}{30}$$

- Diagonals:

$$|V_2V_3| = \frac{\sqrt{2}}{6} \text{ (long diagonal)}, \quad |V_1V_4| = \frac{\sqrt{38}}{35} \text{ (short diagonal)}.$$

The area of a deltoid is half the product of its diagonals:

$$S_{\text{del}} = \frac{1}{2} \cdot |V_2V_3| \cdot |V_1V_4| = \frac{1}{2} \cdot \frac{\sqrt{2}}{6} \cdot \frac{\sqrt{38}}{35} = \frac{\sqrt{76}}{420} = \frac{2\sqrt{19}}{420} = \frac{\sqrt{19}}{210}.$$

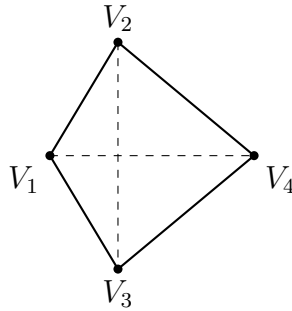


Figure 3: Deltoid — a face of the central deltoidal dodecahedron.

Volume Verification via Pyramids

The center of the polytope has barycentric coordinates $(1/4, 1/4, 1/4, 1/4)$, which in Cartesian coordinates gives

$$O\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right).$$

The plane of the face $x_A = 2x_B$ in Cartesian coordinates is $3x + 3y - z = 2$ (derived from the barycentric relations). The distance from the center to the plane:

$$h = \frac{|3 \cdot \frac{1}{2} + 3 \cdot \frac{1}{2} - \frac{1}{2} - 2|}{\sqrt{3^2 + 3^2 + (-1)^2}} = \frac{|\frac{1}{2}|}{\sqrt{19}} = \frac{1}{2\sqrt{19}}.$$

The volume of the deltoidal dodecahedron, consisting of 12 pyramids with the deltoid base and the vertex at the center, is:

$$V = 12 \cdot \frac{1}{3} S_{\text{del}} h = 4 \cdot \frac{\sqrt{19}}{210} \cdot \frac{1}{2\sqrt{19}} = \frac{4}{420} = \frac{1}{105}.$$

This matches the general formula $\frac{V_{\text{tetra}}}{35} = \frac{1/3}{35} = \frac{1}{105}$, confirming the correctness of the calculations.

The code for the deltoidal dodecahedron is given in Appendix C.

C Asymptote

Code. Central Deltoidal Dodecahedron Inside a Regular Tetrahedron

Three-dimensional images of such polytopes are conveniently created using the open-source language Asymptote [10]. This approach produces high-quality vector interactive images of three-dimensional objects with custom projection and lighting. Below is the code that visualizes the central polytope (deltoidal dodecahedron) obtained in a three-dimensional tetrahedron by trisecting the edges. The reader can copy this code and run it in the Asymptote environment. More information about the language can be found on the official website: <https://asymptote.ualberta.ca/>.

```
import three;
size(10cm);
real d = 0.8;
// All vertices
triple[] V = {
    (4/7, 4/7, 4/7), (0.5, 0.5, 2/3), (0.5, 2/3, 0.5),
    (0.4, 0.6, 0.6), (0.4, 0.4, 0.4), (0.5, 0.5, 1/3),
    (0.5, 1/3, 0.5), (4/7, 3/7, 3/7), (2/3, 0.5, 0.5),
    (0.6, 0.4, 0.6), (1/3, 0.5, 0.5), (3/7, 4/7, 3/7),
    (0.6, 0.6, 0.4), (3/7, 3/7, 4/7)
};
// All 12 faces (deltoids)
int[][] faces = {
    {0,2,12,8}, {0,8,9,1}, {0,1,3,2},
    {1,9,6,13}, {1,13,10,3}, {2,3,10,11},
    {2,11,5,12}, {8,12,5,7}, {8,7,6,9},
    {10,13,6,4}, {10,4,5,11}, {4,6,7,5}};
// Build edges from faces
int[][] edgeSet;
bool hasEdge(int a, int b) {
    for(int[] e : edgeSet) {
        if((e[0] == a && e[1] == b) || (e[0] == b && e[1] == a)) return true;
    }
    return false;
}
for(int[] f : faces) {
    for(int i = 0; i < 4; ++i) {
        int a = f[i];
        int b = f[(i+1)%4];
        if(!hasEdge(a, b)) {
            edgeSet.push(new int[]{a, b});
        }
    }
}
// Draw edges
for(int[] e : edgeSet) {
    if((e[0] == 1 && e[1] == 0) || (e[0] == 0 && e[1] == 1)) {
        draw(V[e[0]] -- V[e[1]], linewidth(0.5));
    } else {
        draw(V[e[0]] -- V[e[1]]);
    }
}
// Fill faces
```

```

for(int[] f : faces) {
path3 poly = V[f[0]] -- V[f[1]] -- V[f[2]] -- V[f[3]] -- cycle;
draw(surface(poly), green+opacity(d));
}
// Projection
currentprojection = perspective(camera=(1.0,2.0,1.5),
up=(0,0,-0),
target=(0.5,0.5,0.5),
zoom=0.,
angle=8.2,
autoadjust=false);

```

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